

1. INTRODUCTION

Building a Text-to-SQL agent that generates SQL from natural language is only half the challenge. The other half is knowing whether it actually works. This document proposes a structured evaluation benchmark and framework to objectively measure the performance of a Text-to-SQL system across multiple dimensions, not just whether the query runs, but whether it returns the right answer.

This framework is designed to be used directly in Tasks 3 and 4 of this assignment series, where the pipeline and agent are tested against the benchmark question dataset.

2. Why Evaluation is Critical

A Text-to-SQL agent can fail in many ways that are not immediately obvious:

- The query runs without error but returns the wrong rows
- The correct table is queried but the wrong column is selected
- A JOIN is missing, causing a cartesian product with inflated row counts
- A WHERE filter is wrong, returning too many or too few results
- The query times out on large tables
- The natural language summary contradicts the actual result

Without a formal evaluation framework, none of these failures would be caught. The goal is to measure all dimensions of correctness: syntax, execution, accuracy, and response quality.

3. Evaluation Dimensions

The framework evaluates the system across 11 dimensions, grouped into 3 tiers:

a. Tier 1: Functional Correctness (Must Pass)

Dimension	What It Measures	How To Check
SQL Syntax Validity	Does the generated SQL parse without error?	Run query: if no exception, pass
Execution Success	Does the query execute on the real database without runtime errors?	Catch exceptions from psycopg2
SELECT only Safety	Does the system reject non-SELECT queries?	Test with DELETE/DROP: should be blocked

b. Tier 2: Result Accuracy (Core Metric)

Dimension	What It Measures	How To Check
Result Accuracy	Does the query return the same data as the manually verified ground truth?	Compare output rows/values to Task 1 Part 1 results
Table Selection	Did the model pick the correct tables?	Parse generated SQL, check table names
Column Selection	Did the model select the right columns?	Parse SELECT clause, compare to expected
JOIN Correctness	Are multi-table join conditions correct?	Check ON clause foreign keys match schema
Filter Accuracy	Are WHERE / GROUP BY / HAVING clauses correct?	Compare result counts to ground truth

c. Tier 3: Agent Intelligence

Dimension	What It Measures	How To Check
Retry Success Rate	When the first attempt fails, does the agent fix and recover?	Count: fixed on retry / total failures
Query Latency	Time from question to final result	time() before and after full pipeline
NL Answer Quality	Is the natural language summary correct and clear?	Human review on 1-5 scale

4. Evaluation Metrics & Targets

Metric	Formula	Target	Priority
Execution Success Rate	$\text{Successful} / \text{Total} \times 100$	> 90%	Critical
Result Accuracy Rate	$\text{Correct Results} / \text{Executed} \times 100$	> 80%	Critical
Table Selection Accuracy	$\text{Correct Tables} / \text{Total} \times 100$	> 90%	High
Column Selection Accuracy	$\text{Correct Cols} / \text{Total} \times 100$	> 85%	High
JOIN Correctness Rate	$\text{Correct Joins} / \text{Join Queries} \times 100$	> 85%	High
Retry Success Rate	$\text{Fixed on Retry} / \text{Failures} \times 100$	> 70%	Medium
Avg Query Latency	$\text{Sum}(\text{latency}) / \text{Total}$	< 5s	Medium
NL Answer Quality	Avg human score (1–5)	> 4.0/5	Medium

5. Benchmark Dataset

The evaluation uses the benchmark question dataset provided in assignment. It contains 50 natural language questions covering:

- Simple single table queries (SELECT, WHERE, DISTINCT)
- Aggregation queries (COUNT, SUM, AVG, MAX, MIN with GROUP BY)
- Multi table JOIN queries (2 table and 3 table joins)
- Self join queries (employees → manager relationship)
- LEFT JOIN queries (customers with no orders)

The ground truth SQL for each question was manually written and verified in Task 1, Part 1. These verified queries serve as the reference answers for evaluating agent output.

6. Evaluation Procedure

Step 1 : Run the Agent Against All Questions

Feed every benchmark question into the system one by one. Record: generated SQL, execution result, whether a retry occurred, latency, and NL summary.

Step 2 : Compare Against Ground Truth

For each question, compare the agent's result to the manually verified result from Task 1 Part 1. A result is correct if it returns the same rows and values, the SQL structure does not need to be identical.

Step 3 : Classify Each Result

Status	Meaning
Full Success	Executed correctly, result matches ground truth, no retry needed
Retry Success	First attempt failed, but retry produced correct result
Wrong Result	Executed without error but returned incorrect data
Failed	All attempts exhausted, query never executed successfully
Blocked	Validator correctly blocked a dangerous query

Step 4 : Calculate Scores

After all questions are processed, calculate each metric from Section 4 and record in the evaluation report table.

7. Common Error Categories

Based on analysis of the classicmodels benchmark, errors fall into these categories:

Error Type	Example	Effect on Result	Fix Strategy
Case sensitivity	customerNumber without quotes	Runtime error, column not found	Add double quotes around mixed case columns
Missing JOIN	Querying customerName from orders	Column does not exist error	Add JOIN to the correct table
Wrong filter value	status = 'shipped' (lowercase)	0 rows returned, case mismatch	Match exact string case from schema
Missing GROUP BY col	SELECT name, COUNT(*) GROUP BY	PostgreSQL syntax error	Group by all non-aggregated columns